

WORK EXPERIENCE

CrossRefHVAC · crossrefhvac.com

2026 - Present

- *Founder & Software Engineer, Barrett Technologies LLC*

Python, PostgreSQL/Supabase, Parquet, Next.js, TypeScript, pdfplumber, pytest, Vercel

HVAC/R equipment data platform, launching publicly as SpecTrace. Turns manufacturer documentation and product APIs into cross-referenced, queryable data with per-field provenance, so machine extraction stays auditable at scale. Coverage architecture is sized for an 8M-fact, multi-brand catalog.

- Built a test-driven Python pipeline (378 tests) that extracts structured product data from dense manufacturer PDFs across 100+ change-tracked source documents.
- Mapped an undocumented manufacturer product API with agent-assisted discovery, indexing 27K+ SKUs across 684 configurations and three brand catalogs in the first vendor rollout.
- Built ground-truth verification against an independent product index. Human rulings persist in an adjudication ledger and unverified data cannot ship.
- Designed an OLAP fact-table schema in PostgreSQL, validated against human-curated ground truth and exported as versioned columnar Parquet snapshots.
- Built a human-in-the-loop review UI that overlays extraction results on the source PDF. Each ruling is recorded once and reapplied automatically.
- Serve the dataset through a REST API and an MCP server. Every value carries provenance to its source page, so AI agents can query equipment data with citations.

No Plastic, No Problem. · noplasticnoproblem.com

2023 - Present

- *Founder*

Next.js, React, TypeScript, Tailwind, Sanity CMS, n8n, Claude API, PostHog, Sentry, Supabase, Vercel

Production affiliate content platform serving 10,000+ monthly users with 50+ brand partnerships.

- Built and operate a production affiliate platform serving 10,000+ monthly users, with a Sanity content pipeline and a bidirectional Notion integration powering the editorial workflow.
- Automated SEO brief generation end to end with n8n, DataForSEO, and the Claude API, supporting a 3x increase in content throughput.
- Reduced CDN request volume by 85% and eliminated compute cost spikes by migrating to webhook-driven ISR revalidation.
- Built a custom MCP server connecting Sanity and Notion as a live knowledge graph, giving AI workflows real-time access to article inventory and brand relationships.

- **Haven Technologies**

Fullstack Software Engineer

New York, NY

June 2022 - August 2023

- Redesigned and rebuilt an insurance policy viewer into a scalable SaaS product using Angular, PostgreSQL, Node.js, and Express, with test coverage via Mocha, Chai, and Sinon.
- Implemented NgRx for state management and centralized API call patterns, improving consistency and debuggability across the frontend codebase.

- **Allvue Systems**

Application Engineer, Operations

New York, NY

July 2020 - May 2022

- Resolved TSQL, XML, and XSLT errors in custom portfolio management software, improving reliability for B2B clients.
- Wrote scripts to fix trade data and payment processing inconsistencies, ensuring accuracy in compliance assessments.

EDUCATION

- **Stevens Institute of Technology**

Bachelor of Science in Computer Science

Hoboken, NJ

September 2016 - May 2020

TECHNICAL SKILLS

Programming

TypeScript, JavaScript, Python, SQL, HTML/CSS

Frameworks & Libs.

React, Next.js, Angular, Redux, NgRx, RxJS, Express, Node.js, Tailwind CSS

Tools

Git, GitHub, Vercel, Sanity, n8n, Sentry, PostHog, pytest, Vitest, pdfplumber, Docker, AWS, Jira

Data

PostgreSQL, Supabase, Parquet, MongoDB, Oracle, MySQL